

Colorisation_Correction_Google_Collab

May 22, 2026

```
[8]: from google.colab import drive
drive.mount('/content/gdrive')
```

Drive already mounted at /content/gdrive; to attempt to forcibly remount, call drive.mount("/content/gdrive", force_remount=True).

```
[3]: !ls "/content/gdrive"
```

MyDrive

```
[4]: import sys
!mkdir -p "/content/gdrive/MyDrive/Colab Notebooks/ML_FDS"
my_local_drive='/content/gdrive/MyDrive/Colab Notebooks/ML_FDS'
sys.path.append(my_local_drive)
```

```
[5]: !ls "/content/gdrive/MyDrive/Colab Notebooks"
```

Colorisation_Correction_Google_Collab.ipynb Untitled0.ipynb
ML_FDS Untitled1.ipynb
Sheep_Cat_Elephant.ipynb

```
[6]: cd $my_local_drive
```

/content/gdrive/MyDrive/Colab Notebooks/ML_FDS

```
[9]: import warnings # Permet de gérer les avertissements
warnings.filterwarnings("ignore", category=FutureWarning)

# Librairies générales
import pandas as pd # Manipulation de données en format tabulaire
from scipy.stats import randint # Pour les distributions aléatoires
import numpy as np # Calcul numérique
from numpy import mean, std # Statistiques de base
import os # Interaction avec le système d'exploitation
from os import listdir # Liste les fichiers dans un répertoire
from os.path import isfile, join # Fonctions pour gérer les chemins de fichiers
import sys # Manipulation du système
import random # Génération de nombres aléatoires
```

```

import cv2 # Librairie pour la vision par ordinateur
import zipfile # Manipulation des archives ZIP
import struct
import tarfile
import urllib.request
from os.path import join
from pathlib import Path

# Librairie d'affichage
import matplotlib.pyplot as plt # Visualisation de données
from PIL import Image, ImageDraw, ImageFilter, ImageEnhance # Manipulation
↳ d'images avec Pillow

# Scikit-learn pour l'apprentissage automatique
from sklearn.decomposition import PCA # Analyse en composantes principales (ACP)

# TensorFlow et Keras pour le deep learning
import tensorflow as tf # Framework principal pour l'apprentissage profond
import keras # API de haut niveau pour TensorFlow
from keras import layers # Modules de construction de modèles Keras
from keras import models # Définition et manipulation des modèles
from keras import optimizers # Optimisateurs pour les modèles Keras
from keras.optimizers import Adam # Optimiseur Adam
from tensorflow.keras.preprocessing.image import ImageDataGenerator # Générateur

# Utilitaires Keras pour le traitement d'images
from keras.utils import img_to_array, load_img # Conversion images en tableaux
from keras.callbacks import (ModelCheckpoint,
                             EarlyStopping, ReduceLROnPlateau) # Gestion des
↳ checkpoints
from keras import backend as K # Backend Keras pour opérations de bas niveau
from keras.models import Model, load_model # Définition/chargement des modèles
from keras.layers import LeakyReLU # Activation LeakyReLU
from keras.preprocessing import image # Prétraitement des images

# Couches Keras pour les modèles convolutifs et autres
from keras.layers import (Input, Conv2D, Flatten, Dense, Conv2DTranspose,
                          Reshape, Activation,
                          BatchNormalization,
                          Dropout, Concatenate ) # Couches pour CNN et
↳ autoencoders
from keras.layers import (InputLayer, MaxPooling2D,
                          UpSampling2D)

```

```

[10]: import shutil
# Nom local de l'archive et URL source
zip_file = "Humans.zip"

```

```

zip_url = "https://www.lirmm.fr/~poncelet/Ressources/Humans.zip"

# Répertoire cible après extraction
src_dir = "Humans"
image_dir = os.path.join(src_dir, "train")

# Vérifie si le dossier de destination contient des images
def humans_already_present():
    return os.path.exists(image_dir) and any(f.endswith(".png") for f in os.
↳listdir(image_dir))

# Téléchargement et extraction seulement si besoin
if not humans_already_present():
    print("Aucune image trouvée dans Humans/train. Téléchargement et extraction_
↳en cours...")

    # Télécharger si l'archive n'est pas déjà présente
    if not os.path.exists(zip_file):
        print(f"Téléchargement de {zip_file}...")
        !wget -O {zip_file} {zip_url}
    else:
        print(f"{zip_file} déjà présent, téléchargement ignoré.")

    # Extraction dans le répertoire courant
    with zipfile.ZipFile(zip_file, "r") as zip_ref:
        for member in zip_ref.namelist():
            if not member.startswith("__MACOSX"):
                zip_ref.extract(member, ".")

    print("Archive extraite dans le répertoire courant.")

    # Suppression de l'archive après extraction
    os.remove(zip_file)
else:
    print("Le dossier Humans/train existe déjà et contient des images.
↳Téléchargement ignoré.")

```

Le dossier Humans/train existe déjà et contient des images. Téléchargement ignoré.

```

[11]: HUMANS_DIR = "Humans/train"
      STL10_DIR  = 'STL10/train'          # sera créé
      COMBINED_DIRS = [HUMANS_DIR, STL10_DIR] # les deux ensemble

      TARGET_SIZE = (128, 128)
      BATCH_SIZE  = 32

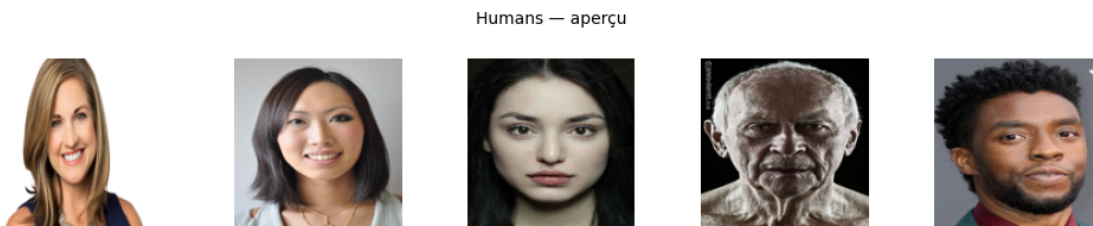
```

```
os.makedirs(STL10_DIR, exist_ok=True)
```

```
[12]: humans_files = [f for f in os.listdir(HUMANS_DIR) if f.endswith('.png')]
print(f'Humans/train : {len(humans_files)} images')

# Aperçu rapide
sample = random.sample(humans_files, 5)
fig, axes = plt.subplots(1, 5, figsize=(10, 2))
for ax, fname in zip(axes, sample):
    ax.imshow(Image.open(join(HUMANS_DIR, fname)))
    ax.axis('off')
plt.suptitle('Humans - aperçu', fontsize=10)
plt.tight_layout()
plt.show()
```

Humans/train : 6834 images



```
[13]: STL10_URL      = 'http://ai.stanford.edu/~acoates/stl10/stl10_binary.tar.gz'
STL10_ARCHIVE = 'stl10_binary.tar.gz'
STL10_N       = 12_000 # combien d'images on extrait du split unlabeled (max
↳ 100 000)

def download_stl10():
    if os.path.exists(STL10_ARCHIVE):
        print("archive déjà présente")
        return

    print("téléchargement STL-10")

    urllib.request.urlretrieve(STL10_URL, STL10_ARCHIVE)

    if os.path.getsize(STL10_ARCHIVE) < 1e8:
        raise ValueError("Téléchargement incomplet : fichier cassé")

    print("OK téléchargement terminé")
```

```

def extract_stl10_unlabeled(n=STL10_N):
    """
    Extrait n images du split 'unlabeled' de STL-10 vers STL10_DIR.

    Format binaire STL-10 :
    - Chaque image = 27 648 octets (3 canaux × 96 × 96)
    - Stockage channel-first : tous les R, puis tous les G, puis tous les B
    - Pas d'en-tête, juste une séquence d'images bout à bout

    On redimensionne à 128×128 en Lanczos (meilleure qualité que BILINEAR
    pour un upscaling de 96→128).
    """
    existing = [f for f in os.listdir(STL10_DIR) if f.endswith('.png')]
    if len(existing) >= n:
        print(f'STL10/train contient déjà {len(existing)} images, extraction_
↳ ignorée.')
        return

    print(f'Extraction de {n} images depuis l\'archive...')

    with tarfile.open(STL10_ARCHIVE, 'r:gz') as tar:
        # Le fichier unlabeled est à ce chemin dans l'archive
        member = tar.getmember('stl10_binary/unlabeled_X.bin')
        f = tar.extractfile(member)

        IMAGE_BYTES = 3 * 96 * 96 # 27 648 octets par image
        saved = 0

        while saved < n:
            raw = f.read(IMAGE_BYTES)
            if len(raw) < IMAGE_BYTES:
                break # fin du fichier (ne devrait pas arriver avant 100 000)

            # Reshape channel-first → channel-last
            # (3, 96, 96) → transpose → (96, 96, 3)
            arr = np.frombuffer(raw, dtype=np.uint8)
            arr = arr.reshape(3, 96, 96).transpose(1, 2, 0) # channel-last

            img = Image.fromarray(arr, mode='RGB')
            img = img.resize(TARGET_SIZE, Image.LANCZOS) # 96→128

            img.save(join(STL10_DIR, f'stl10_{saved:06d}.png'))
            saved += 1

        if saved % 1000 == 0:

```

```

        print(f'  {saved}/{n} images extraites')

    print(f'Extraction terminée : {saved} images dans {STL10_DIR}')

# Exécution
download_stl10()
extract_stl10_unlabeled(n=STL10_N)

```

archive déjà présente

STL10/train contient déjà 12000 images, extraction ignorée.

```

[14]: # Vérification et aperçu
stl_files = [f for f in os.listdir(STL10_DIR) if f.endswith('.png')]
print(f'STL10/train : {len(stl_files)} images')

sample = random.sample(stl_files, 8)
fig, axes = plt.subplots(1, 8, figsize=(16, 2))
for ax, fname in zip(axes, sample):
    ax.imshow(Image.open(join(STL10_DIR, fname)))
    ax.axis('off')
plt.suptitle('STL-10 - aperçu (96×96 → 128×128)', fontsize=10)
plt.tight_layout()
plt.show()

# Bilan
total = len(humans_files) + len(stl_files)
print(f'\nTotal combiné : {len(humans_files)} (Humans) + {len(stl_files)}_
↳ (STL-10) = {total} images')

```

STL10/train : 12000 images



Total combiné : 6834 (Humans) + 12000 (STL-10) = 18834 images

```

[15]: def addMask(image: Image.Image) -> np.array :
        """ Ajoute un masque noir de taille aléatoire à une position aléatoire """
        size = random.randint(30,60)
        start = random.randint(0,128-size)
        out = image.copy()

```

```

    ImageDraw.Draw(out).rectangle(
        [(start, start), (start + size, start + size)], fill='black'
    )
    return np.array(out)

def addNoise(image: Image.Image, level: float = 4.0) -> Image.Image:
    """Ajoute du bruit gaussien, l'intensité est varié d'une image à l'autre"""
    scale = 25 * level * random.uniform(0.8, 1.2)
    arr = np.array(image, dtype=np.float32)
    noisy = arr + np.random.normal(0, scale, arr.shape)
    noisy = np.clip(noisy, 0, 255).astype(np.uint8)
    return Image.fromarray(noisy)

def rgbToGreyScale(image: Image.Image) -> np.ndarray:
    """Convertit l'image rgb au niveau de gris"""
    gray = image.convert("L").resize((128, 128))
    arr = np.array(gray)
    return np.stack([arr] * 3, axis=-1)

def addBlur(image: Image.Image, radius: float = None) -> np.ndarray:
    """Applique un flou gaussien """
    if radius is None:
        radius = random.uniform(2.0, 5.0)
    blurred = image.filter(ImageFilter.GaussianBlur(radius=radius))
    return np.array(blurred)

def combined_filters(image: Image.Image) -> np.ndarray:
    """Applique aléatoirement une combinaison de corruptions """
    img = image.copy()
    if random.random() < 0.5:
        img = addNoise(img, level=2.0)
    if random.random() < 0.5:
        img = Image.fromarray(addMask(img))
    if random.random() < 0.3:
        img = Image.fromarray(addBlur(img))
    return np.array(img)

#autres filtres à tester si ça marche avec les autres fonctions :

def add_scratch(image: Image.Image) -> np.ndarray:
    """Ajoute des rayures aléatoires (comme sur une vieille pellicule)"""
    img = np.array(image)
    num_scratch = random.randint(1, 5)
    for _ in range(num_scratch):
        y_start = random.randint(0, img.shape[0])
        x_start = random.randint(0, img.shape[1])

```

```

        length = random.randint(20, 100)
        thickness = random.randint(1, 3)
        angle = random.uniform(0, 2 * np.pi)
        x_end = int(x_start + length * np.cos(angle))
        y_end = int(y_start + length * np.sin(angle))
        cv2.line(img, (x_start, y_start), (x_end, y_end),
                  (255, 255, 255) if random.random() > 0.5 else (0, 0, 0),
                  thickness)
    return img

def add_dust_and_scratches(image: Image.Image) -> np.ndarray:
    """Ajoute de la poussière (petits points blancs/noirs)"""
    img = np.array(image)
    mask = np.random.random(img.shape[:2]) < 0.005 # 0.5% de pixels
    dust_color = np.random.choice([0, 255], size=mask.sum())
    img[mask] = dust_color[:, np.newaxis]
    return img

def add_jpeg_artifacts(image: Image.Image, quality: int = None) -> np.ndarray:
    """Simule une compression JPEG agressive"""
    if quality is None:
        quality = random.randint(10, 60)
    _, encoded = cv2.imencode('.jpg', cv2.cvtColor(np.array(image), cv2.
    COLOR_RGB2BGR),
                                [cv2.IMWRITE_JPEG_QUALITY, quality])
    decoded = cv2.imdecode(encoded, cv2.IMREAD_COLOR)
    return cv2.cvtColor(decoded, cv2.COLOR_BGR2RGB)

def add_pixelation(image: Image.Image, block_size: int = None) -> np.ndarray:
    """Effet pixelisé (comme une basse résolution)"""
    if block_size is None:
        block_size = random.randint(4, 16)
    h, w = image.size
    small = image.resize((w // block_size, h // block_size), Image.NEAREST)
    return np.array(small.resize((w, h), Image.NEAREST))

def add_glitch_effect(image: Image.Image) -> np.ndarray:
    """Effet 'glitch' numérique (décalage de bandes)"""
    img = np.array(image)
    h, w = img.shape[:2]
    num_glitches = random.randint(3, 10)

    for _ in range(num_glitches):
        y_start = random.randint(0, h - 20)
        y_end = y_start + random.randint(5, 30)
        shift = random.randint(-30, 30)

```



```

    # Décale une bande horizontale
    glitch_band = img[y_start:y_end, :].copy()
    if shift > 0:
        glitch_band = np.roll(glitch_band, shift, axis=1)
        if shift > 0:
            glitch_band[:, :shift] = 0
    else:
        glitch_band = np.roll(glitch_band, shift, axis=1)
        if shift < 0:
            glitch_band[:, shift:] = 0
    img[y_start:y_end, :] = glitch_band

    return img

#dictionnaire
corruption_funcs = {
    'mask' : lambda img: addMask(img),
    'noise' : lambda img: np.array(addNoise(img)),
    'blur' : lambda img: addBlur(img),
    'blackwhite' : lambda img: rgbToGreyScale(img),
    'combined' : lambda img: combined_filters(img),
}

```

```

[16]: def augment(img: Image.Image) -> Image.Image:
    """
    Augmentation légère appliquée sur l'image originale.
    Paramètre : PIL Image RGB 128×128
    Retour : PIL Image RGB 128×128 augmentée
    """

    # flip horizontal
    if random.random() < 0.5:
        img = img.transpose(Image.FLIP_LEFT_RIGHT)

    # random crop + resize
    if random.random() < 0.3:
        margin = random.randint(6, 18)
        img = img.crop((margin, margin, 128 - margin, 128 - margin))
        img = img.resize(TARGET_SIZE, Image.BILINEAR)

    # saturation
    if random.random() < 0.4:
        factor = random.uniform(0.6, 1.4)
        img = ImageEnhance.Color(img).enhance(factor)

    # luminosité
    if random.random() < 0.3:

```

```

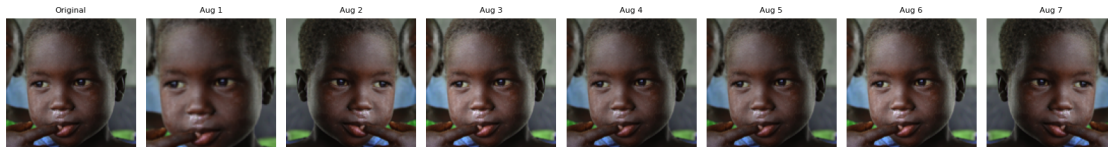
        factor = random.uniform(0.75, 1.25)
        img = ImageEnhance.Brightness(img).enhance(factor)

    return img

# démo
test_path = join(HUMANS_DIR, random.choice(humans_files))
test_img = Image.open(test_path).convert('RGB')

fig, axes = plt.subplots(1, 8, figsize=(16, 2.5))
axes[0].imshow(test_img); axes[0].set_title('Original', fontsize=8); axes[0].axis('off')
for i in range(1, 8):
    axes[i].imshow(augment(test_img))
    axes[i].set_title(f'Aug {i}', fontsize=8)
    axes[i].axis('off')
plt.tight_layout()
plt.show()

```



```

[17]: def tf_augment(img):
    """img : float32 tensor [H, W, 3] in [0,1]"""
    img = tf.image.random_flip_left_right(img)
    if tf.random.uniform(()) < 0.3:
        margin = tf.random.uniform((), 6, 18, dtype=tf.int32)
        crop_size = 128 - margin
        img = tf.image.random_crop(img, [crop_size, crop_size, 3])
        img = tf.image.resize(img, [128, 128])
    img = tf.image.random_saturation(img, 0.6, 1.4)
    img = tf.image.random_brightness(img, 0.25)
    img = tf.clip_by_value(img, 0.0, 1.0)
    return img

```

```

[18]: def tf_corrupt(img, typeimage):
    """img : float32 [H,W,3] in [0,1]"""
    if typeimage == 'blackwhite':
        gray = tf.image.rgb_to_grayscale(img)
        return tf.repeat(gray, 3, axis=-1)

```

```

elif typeimage == 'noise':
    scale = tf.random.uniform(), 0.05, 0.15)
    noise = tf.random.normal(tf.shape(img), stddev=scale)
    return tf.clip_by_value(img + noise, 0.0, 1.0)

elif typeimage == 'blur':
    k = tf.random.uniform(), 2, 5, dtype=tf.int32)
    x = img[tf.newaxis]
    x = tf.nn.avg_pool2d(x, ksize=k, strides=1, padding='SAME')
    return x[0]

elif typeimage == 'mask':
    size = tf.random.uniform(), 30, 60, dtype=tf.int32)
    start = tf.random.uniform(), 0, 128 - size, dtype=tf.int32)
    mask = tf.ones([128, 128, 3], dtype=tf.float32)
    zeros = tf.zeros([size, size, 3])
    indices = [[i, j, c]
                for i in range(size) for j in range(size) for c in range(3)]
    pad_before = [start, start, 0]
    pad_after = [128 - start - size, 128 - start - size, 0]
    hole = 1.0 - tf.pad(tf.ones([size, size, 3]),
                        [[start, 128-start-size],
                         [start, 128-start-size], [0,0]])
    return img * hole

elif typeimage == 'combined':
    if tf.random.uniform() < 0.5:
        img = tf_corrupt(img, 'noise')
    if tf.random.uniform() < 0.5:
        img = tf_corrupt(img, 'mask')
    if tf.random.uniform() < 0.3:
        img = tf_corrupt(img, 'blur')
    return img
return img

```

```

[19]: # def make_generator(
#     folders: list,
#     target_size: tuple,
#     batch_size: int,
#     typeimage: str = 'blackwhite',
#     shuffle: bool = True,
#     use_augment: bool = True
# ):
#     """
#     g  n  rateur infini de paires (corrompue, originale) depuis plusieurs
#     dossiers.

```

```

# Différences avec myImageGenerator :
# - Accepte une liste de dossiers
# - applique augment() avant la corruption (train) si use_augment = True
# """
# corrupt_fn = corruption_funcs[typeimage]
# all_files = []
# for folder in folders:
#     all_files += [
#         join(folder, f)
#         for f in os.listdir(folder)
#         if f.endswith('.png')
#     ]

# print(f'[make_generator] {len(all_files)} images au total '
#       f'({typeimage}, augment={use_augment})')

# if shuffle:
#     random.shuffle(all_files)

# idx = 0
# while True:
#     if idx + batch_size > len(all_files):
#         if shuffle:
#             random.shuffle(all_files)
#         idx = 0

#     batch_paths = all_files[idx : idx + batch_size]
#     originals, corrupted = [], []

#     for path in batch_paths:
#         img = Image.open(path).convert('RGB').resize(target_size, Image.
↳BILINEAR)

#         if use_augment:
#             img = augment(img)

#         originals.append(np.array(img) / 255.0)
#         corrupted.append(corrupt_fn(img) / 255.0)

#     idx += batch_size
#     yield (
#         np.array(corrupted, dtype=np.float32),
#         np.array(originals, dtype=np.float32)
#     )

```

```

[21]: def make_tf_dataset(folders, target_size, batch_size,
                        typeimage='blackwhite', use_augment=True):

```

```

all_files = []
for folder in folders:
    all_files += [join(folder, f)
                  for f in os.listdir(folder) if f.endswith('.png')]
print(f'[make_tf_dataset] {len(all_files)} images ({typeimage},  

↪augment={use_augment})')

def load_image(path):
    raw = tf.io.read_file(path)
    img = tf.image.decode_png(raw, channels=3)
    img = tf.image.resize(img, target_size)
    img = tf.cast(img, tf.float32) / 255.0
    return img

def process(img):
    if use_augment:
        img = tf_augment(img)
    orig = img
    corr = tf_corrupt(img, typeimage)
    return corr, orig

ds = tf.data.Dataset.from_tensor_slices(all_files)

# Charger toutes les images cache en RAM (870 Mo en uint8)
ds = ds.map(load_image, num_parallel_calls=2)
#ds = ds.cache()# clé : lecture disque UNE SEULE FOIS
ds = ds.repeat()

if use_augment:
    ds = ds.shuffle(2000)
ds = ds.map(process, num_parallel_calls=2)
ds = ds.batch(batch_size, drop_remainder=True)
ds = ds.prefetch(2)
return ds

```

```

[22]: # test_gen = make_generator(COMBINED_DIRS, TARGET_SIZE, batch_size=8,
#                               typeimage='blackwhite', use_augment=True)
# x_corr, x_orig = next(test_gen)

# print(f'Shape corrupted : {x_corr.shape}')
# print(f'Shape original : {x_orig.shape}')
# print(f'Range [min, max] : [{x_corr.min():.2f}, {x_corr.max():.2f}]')

# fig, axes = plt.subplots(2, 8, figsize=(16, 4))
# for i in range(8):
#     axes[0, i].imshow(x_orig[i])

```

```
# axes[0, i].axis('off')
# axes[1, i].imshow(x_corr[i], cmap='gray')
# axes[1, i].axis('off')
# axes[0, 0].set_ylabel('Originale', fontsize=9)
# axes[1, 0].set_ylabel('N&B', fontsize=9)
# plt.suptitle('Batch test', fontsize=10)
# plt.tight_layout()
# plt.show()
```

```
[23]: n_humans = len([f for f in os.listdir(HUMANS_DIR) if f.endswith('.png')])
n_stl = len([f for f in os.listdir(STL10_DIR) if f.endswith('.png')])
n_total = n_humans + n_stl

steps_train = max(1, int(n_total * 0.8) // BATCH_SIZE)
steps_val = max(1, int(n_total * 0.2) // BATCH_SIZE)

print('=' * 45)
print(f' Humans/train : {n_humans:>6} images')
print(f' STL10/train : {n_stl:>6} images')
print(f' Total brut : {n_total:>6} images')
print(f' Avec augment : ~{n_total * 3:>5} images effectives (x3 diversité)')
print('=' * 45)
print(f' steps_per_epoch (train, 80 %) : {steps_train}')
print(f' validation_steps (val, 20 %) : {steps_val}')
print('=' * 45)
print()
print('Utilisation dans train_task() :')
print()
print(' train_gen = make_generator(COMBINED_DIRS, TARGET_SIZE,')
print('                               BATCH_SIZE, typeimage, use_augment=True)')
print(' val_gen = make_generator(COMBINED_DIRS, TARGET_SIZE,')
print('                               BATCH_SIZE, typeimage, use_augment=False)')
```

```
=====
Humans/train : 6834 images
STL10/train : 12000 images
Total brut : 18834 images
Avec augment : ~56502 images effectives (x3 diversité)
=====
```

```
steps_per_epoch (train, 80 %) : 470
validation_steps (val, 20 %) : 117
=====
```

Utilisation dans train_task() :

```
train_gen = make_generator(COMBINED_DIRS, TARGET_SIZE,
                           BATCH_SIZE, typeimage, use_augment=True)
```

```
val_gen = make_generator(COMBINED_DIRS, TARGET_SIZE,
                        BATCH_SIZE, typeimage, use_augment=False)
```

```
[24]: def l1_loss(y_true, y_pred):
        return tf.reduce_mean(tf.abs(y_true - y_pred))

def mix_loss(alpha=0.84):
    def loss(y_true, y_pred):
        ms = tf.image.ssim(y_true, y_pred, max_val=1.0)
        ms = tf.clip_by_value(ms, 0.0, 1.0)
        return alpha * (1.0 - tf.reduce_mean(ms)) + (1.0 - alpha) * l1_loss(y_true, y_pred)
    loss.__name__ = f'mix_loss_a{int(alpha*100)}'
    return loss

def make_warmup_loss(final_alpha=0.84):
    """
    Retourne (loss_fn, alpha_var) où alpha_var est une tf.Variable.
    """
    # démarre avec l1
    alpha_var = tf.Variable(0.0, trainable=False, dtype=tf.float32, name='ssim_alpha')

    def loss_fn(y_true, y_pred):
        l1 = tf.reduce_mean(tf.abs(y_true - y_pred))

        #partie MS-SSIM: Calculée toujours, sa contribution est contrôlée par alpha_var
        ms_term_val = 1.0 - tf.reduce_mean(
            tf.clip_by_value(
                tf.image.ssim(y_true, y_pred, max_val=1.0),
                0.0, 1.0
            )
        )

        return alpha_var * ms_term_val + (1.0 - alpha_var) * l1

    loss_fn.__name__ = 'warmup_mix_loss'
    return loss_fn, alpha_var

class WarmupLossCallback(tf.keras.callbacks.Callback):
    """
    Modifie alpha_var à la fin de l'époque de warm-up.
    """
    def __init__(self, alpha_var, warmup_epochs, final_alpha=0.84):
        super().__init__()
```

```

self.alpha_var      = alpha_var
self.warmup_epochs  = warmup_epochs
self.final_alpha    = final_alpha
self.switched       = False

def on_epoch_end(self, epoch, logs=None):
    if epoch == self.warmup_epochs - 1 and not self.switched:
        self.alpha_var.assign(self.final_alpha)
        self.switched = True
        print(f'\n[WarmupLoss] époque {epoch+1} : '
              f'alpha 0.0 to {self.final_alpha} (MS-SSIM activ  )')

```

```

[25]: def build_unet_autoencoder(input_shape=(128, 128, 3)):
    def conv_block(x, filters, name):
        x = Conv2D(filters, 3, strides=2, padding='same',
        ↪name=f'{name}_conv')(x)
        x = BatchNormalization(name=f'{name}_bn')(x)
        x = LeakyReLU(name=f'{name}_act')(x)
        return x
    def deconv_block(x, skip, filters, name):
        x = Conv2DTranspose(filters, 3, strides=2, padding='same',
        ↪name=f'{name}_deconv')(x)
        x = BatchNormalization(name=f'{name}_bn1')(x)
        x = LeakyReLU(name=f'{name}_act1')(x)
        x = Concatenate(name=f'{name}_skip')([x, skip])
        x = Conv2D(filters, 3, padding='same', name=f'{name}_merge')(x)
        x = BatchNormalization(name=f'{name}_bn2')(x)
        x = LeakyReLU(name=f'{name}_act2')(x)
        return x
    inputs = Input(shape=input_shape, name='input')
    e1 = conv_block(inputs, 32, 'enc1')
    e2 = conv_block(e1, 64, 'enc2')
    e3 = conv_block(e2, 128, 'enc3')
    e4 = conv_block(e3, 256, 'enc4')
    bn = conv_block(e4, 256, 'bottleneck')
    d4 = deconv_block(bn, e4, 256, 'dec4')
    d3 = deconv_block(d4, e3, 128, 'dec3')
    d2 = deconv_block(d3, e2, 64, 'dec2')
    d1 = deconv_block(d2, e1, 32, 'dec1')
    outputs = Conv2DTranspose(3, 3, strides=2, padding='same',
        ↪activation='sigmoid', name='output')(d1)
    return Model(inputs, outputs, name='unet_autoencoder')

```

```

[26]: def get_fresh_model():
    """
    Cr  e un nouveau mod  le U-Net avec des poids al  atoires.
    """

```



```

tf.keras.backend.clear_session()
model = build_unet_autoencoder()
return model

```

```

[27]: model_dir = './aeplus_models/'
      EPOCHS     = 50
      PATIENCE   = 10
      BATCH_SIZE = 64

```

```

[85]: def train_task(typeimage, model_name, epochs=EPOCHS, warmup=7, final_alpha=0.
      ↪84):
      """
      Entraîne un U-Net propre sur une tâche de restauration

      Paramètres :
      - typeimage : 'blackwhite', 'mask', 'noise', 'blur', 'combined'
      - model_name : nom du fichier .keras à sauvegarder
      - epochs     : nombre max d'époques
      - warmup     : nb d'époques avec L1 seul avant d'activer MS-SSIM
      - alpha      : poids de MS-SSIM dans mix_loss
      """

      print(f'\n{" "*60}')
      print(f'  Tâche : {typeimage} saved in {model_name}, warmup : {warmup} ep.  ␣
      ↪ alpha_final : {final_alpha}')
      print(f'{" "*60}\n')

      model = get_fresh_model()
      # créer la loss et récupérer la var qu'on passera au callback
      loss_fn, alpha_var = make_warmup_loss(final_alpha=final_alpha)
      model.compile(optimizer=Adam(1e-3), loss=loss_fn)

      #n_images     = len([f for f in os.listdir(image_dir) if f.endswith('.png')])
      steps_train = max(1, int(n_total * 0.8) // BATCH_SIZE)
      steps_val   = max(1, int(n_total * 0.2) // BATCH_SIZE)

      train_gen = make_tf_dataset(COMBINED_DIRS, TARGET_SIZE, BATCH_SIZE, ␣
      ↪typeimage, use_augment=True)
      val_gen   = make_tf_dataset(COMBINED_DIRS, TARGET_SIZE, BATCH_SIZE, ␣
      ↪typeimage, use_augment=False)

      model_path = join(model_dir, model_name)

      callbacks = [
          WarmupLossCallback(alpha_var=alpha_var, warmup_epochs=warmup, ␣
      ↪final_alpha=final_alpha),
          EarlyStopping(

```

```

        monitor='val_loss', patience=PATIENCE,
        restore_best_weights=True, verbose=1),
    ModelCheckpoint(
        filepath=model_path, monitor='val_loss',
        save_best_only=True, verbose=0 ),
    ReduceLROnPlateau(
        monitor='val_loss', factor=0.5,
        patience=4, min_lr=1e-6, verbose=1),
]

history = model.fit(
    train_gen,
    validation_data=val_gen,
    epochs=epochs,
    callbacks=callbacks,
    verbose=1
)

plot_history(history, typeimage)
return model, history

def plot_history(history, title=''):
    """Courbe d'apprentissage avec annotation du point de switch loss."""
    fig, ax = plt.subplots(figsize=(9, 4))
    ax.plot(history.history['loss'], label='Entraînement', lw=2)
    ax.plot(history.history['val_loss'], label='Validation', lw=2,
    ↪linestyle='--')
    ax.set_xlabel('Époque')
    ax.set_ylabel('Perte')
    ax.set_title(f"Courbe d'apprentissage {title}")
    ax.legend()
    ax.grid(alpha=0.3)
    plt.tight_layout()
    plt.show()

```

```

[86]: model_bw, history_bw = train_task(
    typeimage = 'blackwhite',
    model_name = 'UNET_bw_combined.keras',
    epochs     = 50,
    warmup     = 10,
    final_alpha = 0.84
)

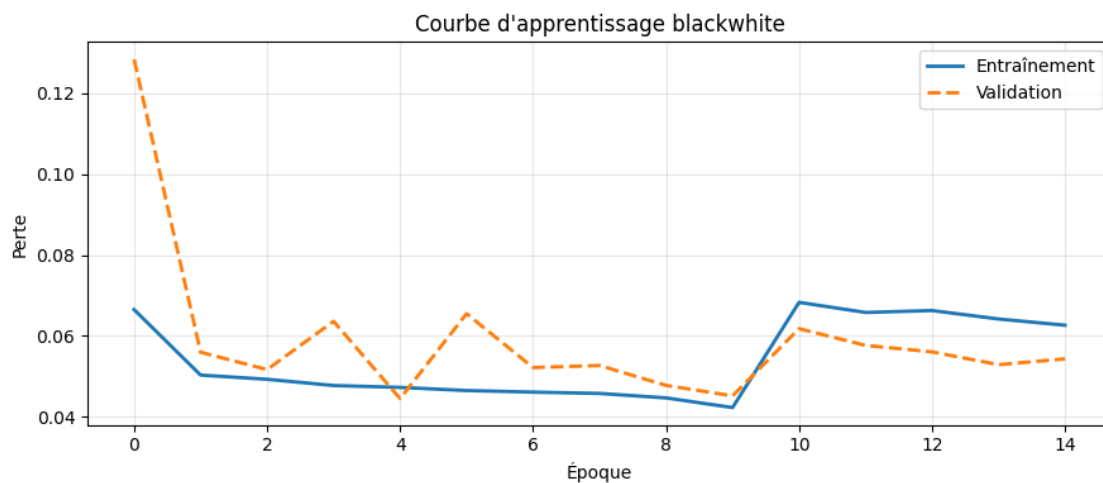
```

Tâche : blackwhite saved in unet_bw_combined.keras, warmup : 10 ep.

alpha_final : 0.84

```
[make_tf_dataset] 18834 images (blackwhite, augment=True)
[make_tf_dataset] 18834 images (blackwhite, augment=False)
Epoch 1/50
294/294          228s 319ms/step -
loss: 0.0665 - val_loss: 0.1284 - learning_rate: 0.0010
Epoch 2/50
294/294          36s 121ms/step -
loss: 0.0503 - val_loss: 0.0559 - learning_rate: 0.0010
Epoch 3/50
294/294          50s 153ms/step -
loss: 0.0492 - val_loss: 0.0516 - learning_rate: 0.0010
Epoch 4/50
294/294          34s 116ms/step -
loss: 0.0477 - val_loss: 0.0636 - learning_rate: 0.0010
Epoch 5/50
294/294          41s 117ms/step -
loss: 0.0472 - val_loss: 0.0444 - learning_rate: 0.0010
Epoch 6/50
294/294          34s 117ms/step -
loss: 0.0464 - val_loss: 0.0655 - learning_rate: 0.0010
Epoch 7/50
294/294          35s 118ms/step -
loss: 0.0460 - val_loss: 0.0521 - learning_rate: 0.0010
Epoch 8/50
294/294          41s 119ms/step -
loss: 0.0457 - val_loss: 0.0526 - learning_rate: 0.0010
Epoch 9/50
294/294          0s 81ms/step -
loss: 0.0455
Epoch 9: ReduceLROnPlateau reducing learning rate to 0.0005000000237487257.
294/294          34s 117ms/step -
loss: 0.0446 - val_loss: 0.0477 - learning_rate: 0.0010
Epoch 10/50
294/294          0s 82ms/step -
loss: 0.0425
[WarmupLoss] époque 10 : alpha 0.0 to 0.84 (MS-SSIM activé)
294/294          33s 113ms/step -
loss: 0.0422 - val_loss: 0.0451 - learning_rate: 5.0000e-04
Epoch 11/50
294/294          35s 120ms/step -
loss: 0.0683 - val_loss: 0.0618 - learning_rate: 5.0000e-04
Epoch 12/50
294/294          35s 118ms/step -
loss: 0.0658 - val_loss: 0.0576 - learning_rate: 5.0000e-04
Epoch 13/50
```

294/294 0s 83ms/step -
 loss: 0.0661
 Epoch 13: ReduceLROnPlateau reducing learning rate to 0.0002500000118743628.
 294/294 33s 113ms/step -
 loss: 0.0662 - val_loss: 0.0560 - learning_rate: 5.0000e-04
 Epoch 14/50
 294/294 35s 119ms/step -
 loss: 0.0641 - val_loss: 0.0528 - learning_rate: 2.5000e-04
 Epoch 15/50
 294/294 44s 151ms/step -
 loss: 0.0626 - val_loss: 0.0543 - learning_rate: 2.5000e-04
 Epoch 15: early stopping
 Restoring model weights from the end of the best epoch: 5.



```
[90]: def compute_psnr(y_true, y_pred):
    return tf.reduce_mean(tf.image.psnr(y_true, y_pred, max_val=1.0))

def show_reconstruction(model, typeimage, n=6):
    gen = make_generator(COMBINED_DIRS, TARGET_SIZE,
                        batch_size=n, typeimage=typeimage,
                        use_augment=False)

    corrupted, orig = next(gen)
    recon = model.predict(corrupted, verbose=0)

    fig, axes = plt.subplots(3, n, figsize=(3 * n, 9))
    for i in range(n):
        score = compute_psnr(orig[i], recon[i]).numpy()
        color = 'green' if score > 25 else 'firebrick'
        cmap = 'gray' if typeimage == 'blackwhite' else None

        axes[0, i].imshow(np.clip(corrupted[i], 0, 1), cmap=cmap)
```

```

axes[0, i].axis('off')
axes[0, i].set_title(f'Sample {i+1}', fontsize=9)

axes[1, i].imshow(np.clip(recon[i], 0, 1))
axes[1, i].axis('off')
axes[1, i].set_xlabel(f'PSNR {score:.1f} dB', color=color, fontsize=8)

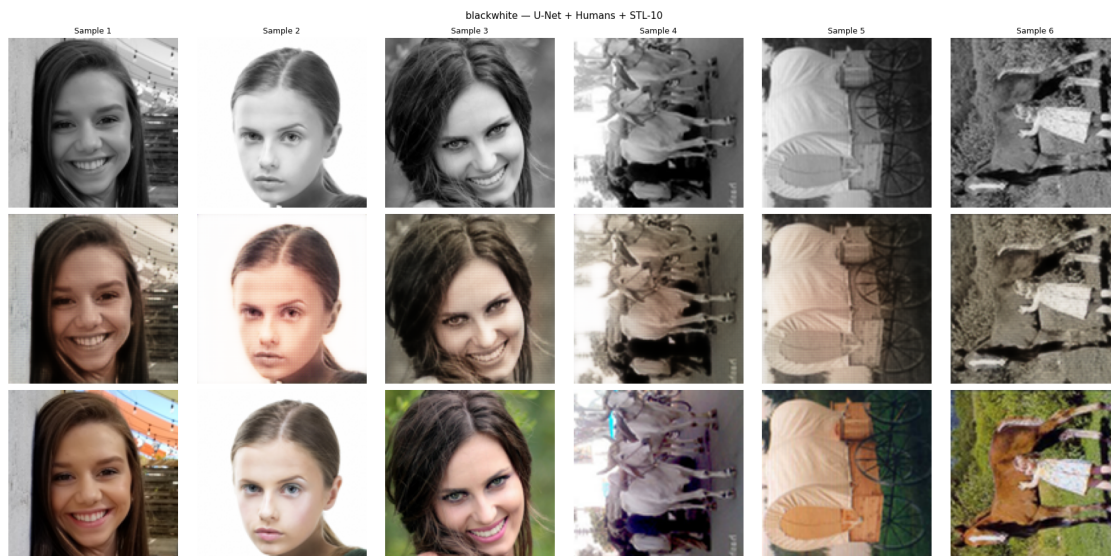
axes[2, i].imshow(np.clip(orig[i], 0, 1))
axes[2, i].axis('off')

axes[0, 0].set_ylabel('N&B',          fontsize=11)
axes[1, 0].set_ylabel('Reconstruite',  fontsize=11)
axes[2, 0].set_ylabel('Originale',     fontsize=11)
plt.suptitle(f'{typeimage} - U-Net + Humans + STL-10', fontsize=11)
plt.tight_layout()
plt.show()

show_reconstruction(model_bw, 'blackwhite')

```

[make_generator] 18834 images au total (blackwhite, augment=False)



on remarque des teintes assez bonne pour quelque samples, mais on remarque that it fails at capturing bright colors, or just pretty much hallucinating some other colors. the desaturation is to be expected because it averages over the values (l1).

the massive issue is our loss graph, the model has been saved before applying ms-ssim. the model did not adapt to our alpha passing from 0.0 to 0.84 so that's what we will fix.

```
[24]: class WarmupLossCallback(tf.keras.callbacks.Callback):
    """
    Montée progressive de alpha sur plusieurs époques après le warmup
    """
    def __init__(self, alpha_var, warmup_epochs, ramp_epochs=5, final_alpha=0.
↪84):
        super().__init__()
        self.alpha_var = alpha_var
        self.warmup_epochs = warmup_epochs
        self.ramp_epochs = ramp_epochs
        self.final_alpha = final_alpha

    def on_epoch_end(self, epoch, logs=None):
        e = epoch + 1
        if e < self.warmup_epochs:
            pass # L1 pur
        elif e < self.warmup_epochs + self.ramp_epochs:
            progress = (e - self.warmup_epochs) / self.ramp_epochs
            new_alpha = self.final_alpha * progress
            self.alpha_var.assign(new_alpha)
            print(f'\n[WarmupLoss] époque {e} : alpha → {new_alpha:.3f}')
        else:
            self.alpha_var.assign(self.final_alpha)
```

```
[24]: def train_task(typeimage, model_name, epochs=EPOCHS, warmup=7, final_alpha=0.
↪84):
    """
    Entraîne un U-Net propre sur une tâche de restauration

    Paramètres :
    - typeimage : 'blackwhite', 'mask', 'noise', 'blur', 'combined'
    - model_name : nom du fichier .keras à sauvegarder
    - epochs : nombre max d'époques
    - warmup : nb d'époques avec L1 seul avant d'activer MS-SSIM
    - alpha : poids de MS-SSIM dans mix_loss
    """
    print(f'\n{" "*60}')
    print(f' Tâche : {typeimage} saved in {model_name}, warmup : {warmup} ep.  ␣
↪ alpha_final : {final_alpha}')
    print(f'{" "*60}\n')

    model = get_fresh_model()
    # créer la loss et récupérer la var qu'on passera au callback
    loss_fn, alpha_var = make_warmup_loss(final_alpha=final_alpha)
    model.compile(optimizer=Adam(1e-3), loss=loss_fn)

    #n_images = len([f for f in os.listdir(image_dir) if f.endswith('.png')])
```

```

steps_train = max(1, int(n_total * 0.8) // BATCH_SIZE)
steps_val    = max(1, int(n_total * 0.2) // BATCH_SIZE)

train_gen = make_tf_dataset(COMBINED_DIRS, TARGET_SIZE, BATCH_SIZE,
↪typeimage, use_augment=True)
val_gen    = make_tf_dataset(COMBINED_DIRS, TARGET_SIZE, BATCH_SIZE,
↪typeimage, use_augment=False)

model_path = join(model_dir, model_name)

callbacks = [
    WarmupLossCallback(alpha_var=alpha_var,
↪warmup_epochs=warmup, ramp_epochs=5, final_alpha=final_alpha),
    EarlyStopping(
        monitor='val_loss', patience=PATIENCE,
        restore_best_weights=True, start_from_epoch=warmup + 5, verbose=1),
    ModelCheckpoint(
        filepath=model_path, monitor='val_loss',
        save_best_only=True, verbose=0 ),
    ReduceLROnPlateau(
        monitor='val_loss', factor=0.5,
        patience=4, min_lr=1e-6, verbose=1),
]

history = model.fit(
    train_gen,
    validation_data=val_gen,
    epochs=epochs,
    callbacks=callbacks,
    verbose=1
)

plot_history(history, typeimage)
return model, history

def plot_history(history, title=''):
    """Courbe d'apprentissage avec annotation du point de switch loss."""
    fig, ax = plt.subplots(figsize=(9, 4))
    ax.plot(history.history['loss'],      label='Entraînement', lw=2)
    ax.plot(history.history['val_loss'],  label='Validation',   lw=2,
↪linestyle='--')
    ax.set_xlabel('Époque')
    ax.set_ylabel('Perte')
    ax.set_title(f"Courbe d'apprentissage {title}")
    ax.legend()
    ax.grid(alpha=0.3)

```

```
plt.tight_layout()
plt.show()
```

```
[25]: model_bw, history_bw = train_task(
        typeimage = 'blackwhite',
        model_name = 'unet_bw_combined_2.keras',
        epochs     = 50,
        warmup      = 10,
        final_alpha = 0.84
    )
```

Tâche : blackwhite saved in unet_bw_combined_2.keras, warmup : 10 ep.
alpha_final : 0.84

```
[make_tf_dataset] 18834 images (blackwhite, augment=True)
[make_tf_dataset] 18834 images (blackwhite, augment=False)
Epoch 1/50
294/294          241s 325ms/step -
loss: 0.0673 - val_loss: 0.1504 - learning_rate: 0.0010
Epoch 2/50
294/294          36s 121ms/step -
loss: 0.0505 - val_loss: 0.0534 - learning_rate: 0.0010
Epoch 3/50
294/294          35s 118ms/step -
loss: 0.0475 - val_loss: 0.0469 - learning_rate: 0.0010
Epoch 4/50
294/294          35s 119ms/step -
loss: 0.0471 - val_loss: 0.0450 - learning_rate: 0.0010
Epoch 5/50
294/294          40s 116ms/step -
loss: 0.0461 - val_loss: 0.0786 - learning_rate: 0.0010
Epoch 6/50
294/294          35s 120ms/step -
loss: 0.0465 - val_loss: 0.0561 - learning_rate: 0.0010
Epoch 7/50
294/294          34s 117ms/step -
loss: 0.0450 - val_loss: 0.0534 - learning_rate: 0.0010
Epoch 8/50
294/294           0s 83ms/step -
loss: 0.0446
Epoch 8: ReduceLROnPlateau reducing learning rate to 0.0005000000237487257.
294/294          35s 118ms/step -
loss: 0.0450 - val_loss: 0.0455 - learning_rate: 0.0010
Epoch 9/50
```



```

294/294          35s 119ms/step -
loss: 0.0426 - val_loss: 0.0436 - learning_rate: 5.0000e-04
Epoch 10/50
294/294          0s 83ms/step -
loss: 0.0416
[WarmupLoss] époque 10 : alpha → 0.000
294/294          34s 114ms/step -
loss: 0.0413 - val_loss: 0.0467 - learning_rate: 5.0000e-04
Epoch 11/50
294/294          0s 83ms/step -
loss: 0.0410
[WarmupLoss] époque 11 : alpha → 0.168
294/294          43s 120ms/step -
loss: 0.0409 - val_loss: 0.0406 - learning_rate: 5.0000e-04
Epoch 12/50
294/294          0s 85ms/step -
loss: 0.0459
[WarmupLoss] époque 12 : alpha → 0.336
294/294          35s 120ms/step -
loss: 0.0457 - val_loss: 0.0439 - learning_rate: 5.0000e-04
Epoch 13/50
294/294          0s 83ms/step -
loss: 0.0508
[WarmupLoss] époque 13 : alpha → 0.504
294/294          35s 118ms/step -
loss: 0.0505 - val_loss: 0.0487 - learning_rate: 5.0000e-04
Epoch 14/50
294/294          0s 83ms/step -
loss: 0.0552
[WarmupLoss] époque 14 : alpha → 0.672
294/294          35s 119ms/step -
loss: 0.0552 - val_loss: 0.0491 - learning_rate: 5.0000e-04
Epoch 15/50
294/294          0s 82ms/step -
loss: 0.0594
Epoch 15: ReduceLROnPlateau reducing learning rate to 0.0002500000118743628.
294/294          34s 115ms/step -
loss: 0.0590 - val_loss: 0.0531 - learning_rate: 5.0000e-04
Epoch 16/50
294/294          34s 117ms/step -
loss: 0.0625 - val_loss: 0.0518 - learning_rate: 2.5000e-04
Epoch 17/50
294/294          35s 118ms/step -
loss: 0.0618 - val_loss: 0.0523 - learning_rate: 2.5000e-04
Epoch 18/50
294/294          33s 113ms/step -
loss: 0.0621 - val_loss: 0.0523 - learning_rate: 2.5000e-04
Epoch 19/50

```

```

294/294          0s 83ms/step -
loss: 0.0616
Epoch 19: ReduceLROnPlateau reducing learning rate to 0.0001250000059371814.
294/294          35s 118ms/step -
loss: 0.0619 - val_loss: 0.0514 - learning_rate: 2.5000e-04
Epoch 20/50
294/294          34s 116ms/step -
loss: 0.0611 - val_loss: 0.0517 - learning_rate: 1.2500e-04
Epoch 21/50
294/294          35s 118ms/step -
loss: 0.0612 - val_loss: 0.0510 - learning_rate: 1.2500e-04
Epoch 22/50
294/294          34s 117ms/step -
loss: 0.0609 - val_loss: 0.0522 - learning_rate: 1.2500e-04
Epoch 23/50
294/294          0s 82ms/step -
loss: 0.0609
Epoch 23: ReduceLROnPlateau reducing learning rate to 6.25000029685907e-05.
294/294          35s 117ms/step -
loss: 0.0607 - val_loss: 0.0511 - learning_rate: 1.2500e-04
Epoch 24/50
294/294          41s 118ms/step -
loss: 0.0603 - val_loss: 0.0505 - learning_rate: 6.2500e-05
Epoch 25/50
294/294          35s 118ms/step -
loss: 0.0604 - val_loss: 0.0506 - learning_rate: 6.2500e-05
Epoch 26/50
294/294          41s 117ms/step -
loss: 0.0603 - val_loss: 0.0506 - learning_rate: 6.2500e-05
Epoch 27/50
294/294          0s 83ms/step -
loss: 0.0600
Epoch 27: ReduceLROnPlateau reducing learning rate to 3.125000148429535e-05.
294/294          41s 118ms/step -
loss: 0.0602 - val_loss: 0.0505 - learning_rate: 6.2500e-05
Epoch 28/50
294/294          39s 112ms/step -
loss: 0.0603 - val_loss: 0.0502 - learning_rate: 3.1250e-05
Epoch 29/50
294/294          41s 112ms/step -
loss: 0.0600 - val_loss: 0.0505 - learning_rate: 3.1250e-05
Epoch 30/50
294/294          34s 117ms/step -
loss: 0.0600 - val_loss: 0.0504 - learning_rate: 3.1250e-05
Epoch 31/50
294/294          0s 80ms/step -
loss: 0.0607
Epoch 31: ReduceLROnPlateau reducing learning rate to 1.5625000742147677e-05.

```

```

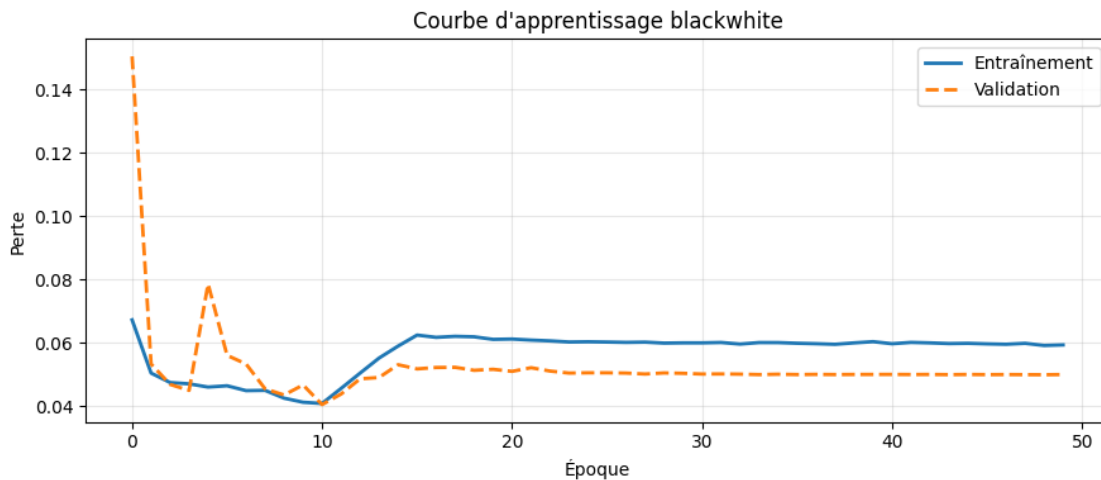
294/294          34s 115ms/step -
loss: 0.0600 - val_loss: 0.0502 - learning_rate: 3.1250e-05
Epoch 32/50
294/294          33s 112ms/step -
loss: 0.0601 - val_loss: 0.0502 - learning_rate: 1.5625e-05
Epoch 33/50
294/294          45s 152ms/step -
loss: 0.0596 - val_loss: 0.0502 - learning_rate: 1.5625e-05
Epoch 34/50
294/294          34s 117ms/step -
loss: 0.0601 - val_loss: 0.0500 - learning_rate: 1.5625e-05
Epoch 35/50
294/294          0s 83ms/step -
loss: 0.0600
Epoch 35: ReduceLROnPlateau reducing learning rate to 7.812500371073838e-06.
294/294          33s 113ms/step -
loss: 0.0601 - val_loss: 0.0501 - learning_rate: 1.5625e-05
Epoch 36/50
294/294          35s 117ms/step -
loss: 0.0599 - val_loss: 0.0500 - learning_rate: 7.8125e-06
Epoch 37/50
294/294          34s 117ms/step -
loss: 0.0598 - val_loss: 0.0501 - learning_rate: 7.8125e-06
Epoch 38/50
294/294          35s 118ms/step -
loss: 0.0596 - val_loss: 0.0500 - learning_rate: 7.8125e-06
Epoch 39/50
294/294          0s 82ms/step -
loss: 0.0607
Epoch 39: ReduceLROnPlateau reducing learning rate to 3.906250185536919e-06.
294/294          41s 118ms/step -
loss: 0.0600 - val_loss: 0.0501 - learning_rate: 7.8125e-06
Epoch 40/50
294/294          34s 117ms/step -
loss: 0.0604 - val_loss: 0.0501 - learning_rate: 3.9063e-06
Epoch 41/50
294/294          33s 112ms/step -
loss: 0.0598 - val_loss: 0.0501 - learning_rate: 3.9063e-06
Epoch 42/50
294/294          34s 116ms/step -
loss: 0.0602 - val_loss: 0.0500 - learning_rate: 3.9063e-06
Epoch 43/50
294/294          0s 81ms/step -
loss: 0.0602
Epoch 43: ReduceLROnPlateau reducing learning rate to 1.9531250927684596e-06.
294/294          34s 116ms/step -
loss: 0.0600 - val_loss: 0.0501 - learning_rate: 3.9063e-06
Epoch 44/50

```

```

294/294          33s 111ms/step -
loss: 0.0598 - val_loss: 0.0500 - learning_rate: 1.9531e-06
Epoch 45/50
294/294          35s 117ms/step -
loss: 0.0599 - val_loss: 0.0501 - learning_rate: 1.9531e-06
Epoch 46/50
294/294          40s 114ms/step -
loss: 0.0597 - val_loss: 0.0500 - learning_rate: 1.9531e-06
Epoch 47/50
294/294          0s 82ms/step -
loss: 0.0596
Epoch 47: ReduceLROnPlateau reducing learning rate to 1e-06.
294/294          35s 117ms/step -
loss: 0.0596 - val_loss: 0.0500 - learning_rate: 1.9531e-06
Epoch 48/50
294/294          33s 113ms/step -
loss: 0.0599 - val_loss: 0.0500 - learning_rate: 1.0000e-06
Epoch 49/50
294/294          34s 114ms/step -
loss: 0.0592 - val_loss: 0.0500 - learning_rate: 1.0000e-06
Epoch 50/50
294/294          34s 117ms/step -
loss: 0.0594 - val_loss: 0.0500 - learning_rate: 1.0000e-06
Restoring model weights from the end of the best epoch: 49.

```



```

[28]: from skimage.metrics import peak_signal_noise_ratio as compute_psnr
      from skimage.metrics import structural_similarity as compute_ssim

      def show_reconstruction(model, typeimage, n=6, dataset_dirs=COMBINED_DIRS):
          """
          Affiche n triplets : (corrompue / reconstruite / originale)

```

```

avec scores PSNR et SSIM par image.
"""
ds = make_tf_dataset(dataset_dirs, TARGET_SIZE, batch_size=n,
                      typeimage=typeimage, use_augment=False)
corr_batch, orig_batch = next(iter(ds))
corr = corr_batch.numpy()
orig = orig_batch.numpy()
recon = model.predict(corr_batch, verbose=0)
fig, axes = plt.subplots(3, n, figsize=(3.2 * n, 10))

row_labels = {
    0: 'N&B (entrée)' if typeimage == 'blackwhite' else f'{typeimage} (entrée)',
    1: 'Reconstruction',
    2: 'Originale',
}
cmap_input = 'gray' if typeimage == 'blackwhite' else None

for i in range(n):
    orig_u8 = (np.clip(orig[i], 0, 1) * 255).astype(np.uint8)
    recon_u8 = (np.clip(recon[i], 0, 1) * 255).astype(np.uint8)

    psnr = compute_psnr(orig_u8, recon_u8, data_range=255)
    ssim = compute_ssim(orig_u8, recon_u8,
                        data_range=255, channel_axis=-1)

    psnr_color = 'green' if psnr > 25 else ('orange' if psnr > 20 else
    'firebrick')

    # Ligne 0 : image corrompue
    axes[0, i].imshow(np.clip(corr[i], 0, 1), cmap=cmap_input)
    axes[0, i].set_title(f'Sample {i+1}', fontsize=9)
    axes[0, i].axis('off')

    # Ligne 1 : reconstruction
    axes[1, i].imshow(np.clip(recon[i], 0, 1))
    axes[1, i].set_xlabel(
        f'PSNR {psnr:.1f} dB | SSIM {ssim:.3f}',
        color=psnr_color, fontsize=8
    )
    axes[1, i].axis('off')

    # Ligne 2 : originale
    axes[2, i].imshow(np.clip(orig[i], 0, 1))
    axes[2, i].axis('off')

# Labels des lignes

```

```

for row, label in row_labels.items():
    axes[row, 0].set_ylabel(label, fontsize=11, labelpad=8)

plt.suptitle(f'{typeimage} - U-Net (best epoch sauvegardé)', fontsize=12,
↳y=1.01)
plt.tight_layout()
plt.show()

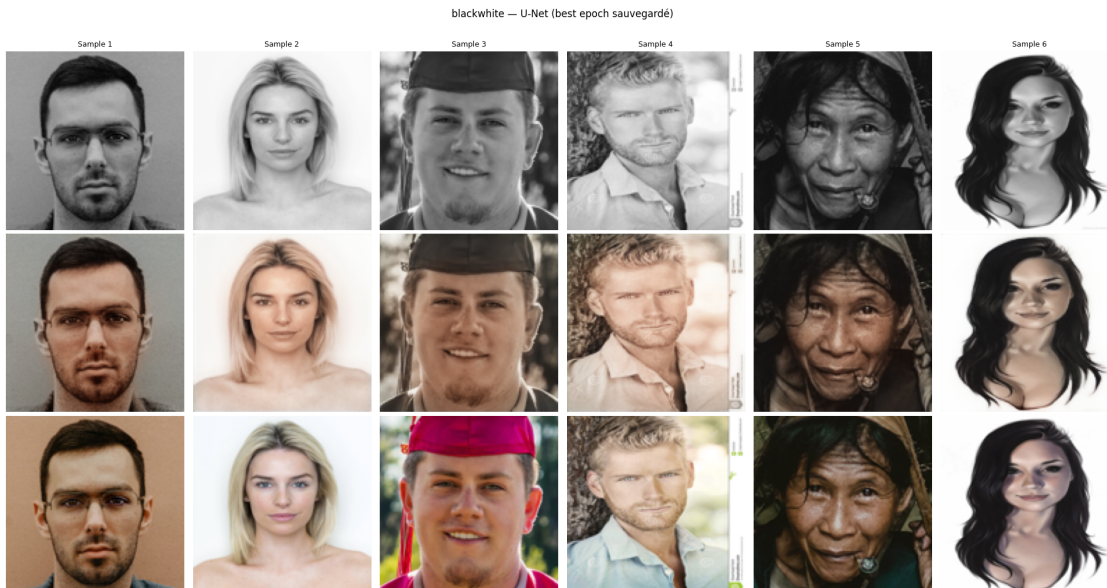
psnr_scores, ssim_scores = [], []
for i in range(n):
    o = (np.clip(orig[i], 0, 1) * 255).astype(np.uint8)
    r = (np.clip(recon[i], 0, 1) * 255).astype(np.uint8)
    psnr_scores.append(compute_psnr(o, r, data_range=255))
    ssim_scores.append(compute_ssim(o, r, data_range=255, channel_axis=-1))

print(f"\n{' '*40}")
print(f" PSNR moyen : {np.mean(psnr_scores):.2f} dB (±{np.
↳std(psnr_scores):.2f})")
print(f" SSIM moyen : {np.mean(ssim_scores):.4f} (±{np.std(ssim_scores):.
↳4f})")
print(f"{' '*40}\n")

show_reconstruction(model_bw, 'blackwhite', n=6)

```

[make_tf_dataset] 18834 images (blackwhite, augment=False)



PSNR moyen : 26.95 dB (± 5.24)
SSIM moyen : 0.9503 (± 0.0533)

Remarques : on remarque que les résultats sont un peu mieux on a également psnr 26.95 > 25 acceptable, et ssim 0.95 qui généralement bon structurellement mais les teintes sont toujours désaturées (orange/brein) qui est un biais de dataset, le modèle se concentre sur les visages mais background est généraliser. pour le graphe de loss :

Epoch 1-10 (L1 pur) : 0.0673 $\rightarrow$ 0.0413 descente propre Epoch 11 (alpha=0.168) : 0.0409 pas de spike Epoch 12 (alpha=0.336) : 0.0457 la loss mssim remet à l'échelle Epoch 13-50 (alpha $\rightarrow$ 0.84) : plateau $\sim$ 0.060 lr descend jusqu'à 1e-6, modèle bloqué Best epoch : 49, val_loss 0.0500

le meilleur checkpoint L1 était epoch 11 (val_loss=0.0406), mais après ça, MS-SSIM pousse la loss dans une autre direction que L1. Les deux métriques ne partagent pas la même échelle, MS-SSIM renvoie des valeurs $\sim 10\times$ plus grandes que L1 pour les mêmes images, donc à alpha=0.84 il domine trop et la val_loss L1-comparable semble monter alors que la qualité perceptuelle s'améliore peut-être.

1 Amélioration des fonctions loss

```
[29]: def pure_l1(y_true, y_pred):
    return tf.reduce_mean(tf.abs(y_true - y_pred))

def lab_color_loss(y_true, y_pred):
    """
    Pénalise davantage les erreurs sur les canaux couleur (A, B)
    que sur la luminance (L).
    """
    # Conversion RGB LAB via une approximation TF
    def rgb_to_lab(img):
        img = tf.where(img > 0.04045,
                        tf.pow((img + 0.055) / 1.055, 2.4),
                        img / 12.92)

        r, g, b = img[...,0:1], img[...,1:2], img[...,2:3]
        x = 0.4124*r + 0.3576*g + 0.1805*b
        y = 0.2126*r + 0.7152*g + 0.0722*b
        z = 0.0193*r + 0.1192*g + 0.9505*b
        xyz = tf.concat([x/0.95047, y/1.00000, z/1.08883], axis=-1)

        eps = 0.008856
        xyz = tf.where(xyz > eps,
                        tf.pow(xyz, 1/3),
                        7.787*xyz + 16/116)

        L = 116*xyz[...,1:2] - 16
        A = 500*(xyz[...,0:1] - xyz[...,1:2])
```

```

        B = 200*(xyz[...,1:2] - xyz[...,2:3])
        return tf.concat([L/100, A/128, B/128], axis=-1)

true_lab = rgb_to_lab(y_true)
pred_lab = rgb_to_lab(y_pred)

l_loss = tf.reduce_mean(tf.abs(true_lab[...,0] - pred_lab[...,0]))
ab_loss = tf.reduce_mean(tf.abs(true_lab[...,1:] - pred_lab[...,1:]))

return l_loss + 3.0 * ab_loss

def ssim_simple(y_true, y_pred, max_val=1.0):
    C1 = (0.01 * max_val) ** 2
    C2 = (0.03 * max_val) ** 2

    def _fspecial_gauss(size, sigma):
        coords = tf.cast(tf.range(size), tf.float32) - (size - 1) / 2.0
        g = tf.exp(-(coords ** 2) / (2 * sigma ** 2))
        g = g / tf.reduce_sum(g)
        kernel = g[:, None] * g[None, :]
        kernel = kernel[:, :, None, None]
        return kernel

    kernel = _fspecial_gauss(11, 1.5)
    kernel = tf.tile(kernel, [1, 1, 3, 1])

    def apply_filter(x):
        return tf.nn.depthwise_conv2d(
            x, kernel, strides=[1,1,1,1], padding='SAME'
        )

    mu1 = apply_filter(y_true)
    mu2 = apply_filter(y_pred)

    mu1_sq = mu1 * mu1
    mu2_sq = mu2 * mu2
    mu1_mu2 = mu1 * mu2

    sigma1_sq = apply_filter(y_true * y_true) - mu1_sq
    sigma2_sq = apply_filter(y_pred * y_pred) - mu2_sq
    sigma12 = apply_filter(y_true * y_pred) - mu1_mu2

    numerator = (2.0 * mu1_mu2 + C1) * (2.0 * sigma12 + C2)
    denominator = (mu1_sq + mu2_sq + C1) * (sigma1_sq + sigma2_sq + C2)

    ssim_map = numerator / (denominator + 1e-8)
    return tf.reduce_mean(ssim_map)

```



```

def make_warmup_loss(final_alpha=0.5):
    alpha_var = tf.Variable(0.0, trainable=False, dtype=tf.float32)

    def loss_fn(y_true, y_pred):
        l1 = tf.reduce_mean(tf.abs(y_true - y_pred))
        lab = lab_color_loss(y_true, y_pred)
        ms = 1.0 - ssim_simple(y_true, y_pred)

        return (1.0 - alpha_var) * (0.5 * l1 + 0.5 * lab) + alpha_var * ms

    loss_fn.__name__ = 'warmup_mix_loss'
    return loss_fn, alpha_var

```

```

[30]: def train_task(typeimage, model_name, epochs=EPOCHS, warmup=7, ramp_epochs=5,
↪ final_alpha=0.5):
    """
    Entraîne un U-Net propre sur une tâche de restauration

    Paramètres :
    - typeimage : 'blackwhite', 'mask', 'noise', 'blur', 'combined'
    - model_name : nom du fichier .keras à sauvegarder
    - epochs : nombre max d'époques
    - warmup : nb d'époques avec L1 seul avant d'activer MS-SSIM
    - ramp_epochs : nombre d'époques pour monter alpha progressivement après le
↪ warmup
    - alpha : poids de MS-SSIM dans mix_loss
    """
    import gc
    tf.keras.backend.clear_session()
    gc.collect()

    print(f'\n{" " * 60}')
    print(f' Tâche : {typeimage} saved in {model_name}, warmup : {warmup} ep. ↪
↪ alpha_final : {final_alpha}')
    print(f'{" " * 60}\n')

    model = get_fresh_model()
    # créer la loss et récupérer la var qu'on passera au callback
    loss_fn, alpha_var = make_warmup_loss(final_alpha=final_alpha)
    model.compile(optimizer=Adam(1e-3), loss=loss_fn, metrics=[pure_l1])

    #n_images = len([f for f in os.listdir(image_dir) if f.endswith('.png')])
    steps_train = max(1, int(n_total * 0.8) // BATCH_SIZE)
    steps_val = max(1, int(n_total * 0.2) // BATCH_SIZE)

```

```

    train_gen = make_tf_dataset(COMBINED_DIRS, TARGET_SIZE, BATCH_SIZE,
↳typeimage, use_augment=True)
    val_gen = make_tf_dataset(COMBINED_DIRS, TARGET_SIZE, BATCH_SIZE,
↳typeimage, use_augment=False)

    callbacks = [
        WarmupLossCallback(alpha_var=alpha_var,
↳warmup_epochs=warmup,ramp_epochs=ramp_epochs, final_alpha=final_alpha),
        EarlyStopping(
            monitor='val_pure_l1', patience=PATIENCE, start_from_epoch= warmup
↳+ ramp_epochs,mode='min',
            restore_best_weights=True, verbose=1),

        ModelCheckpoint(
            filepath=join(model_dir, 'checkpoint_epoch{epoch:02d}.keras'),
            save_freq='epoch', monitor='val_pure_l1',mode='min',
            save_best_only=True, verbose=0 ),

        ReduceLROnPlateau(
            monitor='val_loss', factor=0.5,
            patience=7, min_lr=1e-6, verbose=1),
    ]

    history = model.fit(
        train_gen,
        validation_data=val_gen,
        epochs=epochs,
        steps_per_epoch=steps_train,
        validation_steps=steps_val,
        callbacks=callbacks,
        verbose=1
    )

    plot_history(history, typeimage)
    return model, history

def plot_history(history, title=''):
    """Courbe d'apprentissage avec annotation du point de switch loss."""
    fig, ax = plt.subplots(figsize=(9, 4))
    ax.plot(history.history['loss'], label='Entraînement', lw=2)
    ax.plot(history.history['val_loss'], label='Validation', lw=2,
↳linestyle='--')
    ax.set_xlabel('Époque')
    ax.set_ylabel('Perte')
    ax.set_title(f"Courbe d'apprentissage {title}")
    ax.legend()

```

```
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

```
[31]: # model_bw, history_bw = train_task(
#     typeimage = 'blackwhite',
#     model_name = 'unet_bw_combined_3.keras',
#     epochs     = 30, #because of ram space on google collab
#     warmup     = 10,
#     final_alpha = 0.84
# )
```

Tâche : blackwhite saved in unet_bw_combined_3.keras, warmup : 10 ep.
alpha_final : 0.84

```
[make_tf_dataset] 18834 images (blackwhite, augment=True)
[make_tf_dataset] 18834 images (blackwhite, augment=False)
Epoch 1/30
235/235          174s 318ms/step -
loss: 0.1571 - pure_l1: 0.0777 - val_loss: 0.2733 - val_pure_l1: 0.1803 -
learning_rate: 0.0010
Epoch 2/30
235/235          74s 317ms/step -
loss: 0.1223 - pure_l1: 0.0512 - val_loss: 0.2137 - val_pure_l1: 0.1032 -
learning_rate: 0.0010
Epoch 3/30
235/235          73s 311ms/step -
loss: 0.1188 - pure_l1: 0.0488 - val_loss: 0.1217 - val_pure_l1: 0.0512 -
learning_rate: 0.0010
Epoch 4/30
235/235          71s 303ms/step -
loss: 0.1195 - pure_l1: 0.0483 - val_loss: 0.1184 - val_pure_l1: 0.0482 -
learning_rate: 0.0010
Epoch 5/30
235/235          70s 297ms/step -
loss: 0.1204 - pure_l1: 0.0488 - val_loss: 0.1044 - val_pure_l1: 0.0423 -
learning_rate: 0.0010
Epoch 6/30
235/235          70s 298ms/step -
loss: 0.1151 - pure_l1: 0.0467 - val_loss: 0.1115 - val_pure_l1: 0.0443 -
learning_rate: 0.0010
Epoch 7/30
235/235          68s 291ms/step -
loss: 0.1133 - pure_l1: 0.0459 - val_loss: 0.1238 - val_pure_l1: 0.0519 -
```

```

learning_rate: 0.0010
Epoch 8/30
235/235          82s 350ms/step -
loss: 0.1137 - pure_l1: 0.0460 - val_loss: 0.1272 - val_pure_l1: 0.0544 -
learning_rate: 0.0010
Epoch 9/30
235/235          71s 303ms/step -
loss: 0.1159 - pure_l1: 0.0463 - val_loss: 0.1025 - val_pure_l1: 0.0420 -
learning_rate: 0.0010
Epoch 10/30
235/235          0s 240ms/step -
loss: 0.1095 - pure_l1: 0.0441
[WarmupLoss] époque 10 : alpha → 0.000
235/235          70s 297ms/step -
loss: 0.1160 - pure_l1: 0.0463 - val_loss: 0.1230 - val_pure_l1: 0.0578 -
learning_rate: 0.0010
Epoch 11/30
235/235          0s 243ms/step -
loss: 0.1058 - pure_l1: 0.0430
[WarmupLoss] époque 11 : alpha → 0.168
235/235          81s 347ms/step -
loss: 0.1117 - pure_l1: 0.0447 - val_loss: 0.1514 - val_pure_l1: 0.0553 -
learning_rate: 0.0010
Epoch 12/30
235/235          0s 242ms/step -
loss: 0.1082 - pure_l1: 0.0447
[WarmupLoss] époque 12 : alpha → 0.336
235/235          71s 302ms/step -
loss: 0.1060 - pure_l1: 0.0440 - val_loss: 0.0983 - val_pure_l1: 0.0409 -
learning_rate: 0.0010
Epoch 13/30
235/235          0s 242ms/step -
loss: 0.1044 - pure_l1: 0.0448
[WarmupLoss] époque 13 : alpha → 0.504
235/235          71s 302ms/step -
loss: 0.0981 - pure_l1: 0.0427 - val_loss: 0.0796 - val_pure_l1: 0.0369 -
learning_rate: 0.0010
Epoch 14/30
235/235          0s 241ms/step -
loss: 0.0982 - pure_l1: 0.0440
[WarmupLoss] époque 14 : alpha → 0.672
235/235          70s 298ms/step -
loss: 0.0945 - pure_l1: 0.0426 - val_loss: 0.0753 - val_pure_l1: 0.0393 -
learning_rate: 0.0010
Epoch 15/30
235/235          70s 298ms/step -
loss: 0.0873 - pure_l1: 0.0421 - val_loss: 0.0588 - val_pure_l1: 0.0340 -
learning_rate: 0.0010

```

Epoch 16/30
235/235 70s 299ms/step -
loss: 0.0770 - pure_l1: 0.0407 - val_loss: 0.0683 - val_pure_l1: 0.0445 -
learning_rate: 0.0010
Epoch 17/30
235/235 72s 307ms/step -
loss: 0.0752 - pure_l1: 0.0402 - val_loss: 0.0535 - val_pure_l1: 0.0371 -
learning_rate: 0.0010
Epoch 18/30
235/235 73s 312ms/step -
loss: 0.0746 - pure_l1: 0.0398 - val_loss: 0.0550 - val_pure_l1: 0.0361 -
learning_rate: 0.0010
Epoch 19/30
235/235 81s 345ms/step -
loss: 0.0777 - pure_l1: 0.0413 - val_loss: 0.0521 - val_pure_l1: 0.0343 -
learning_rate: 0.0010
Epoch 20/30
235/235 70s 297ms/step -
loss: 0.0774 - pure_l1: 0.0413 - val_loss: 0.0526 - val_pure_l1: 0.0367 -
learning_rate: 0.0010
Epoch 21/30
235/235 69s 295ms/step -
loss: 0.0734 - pure_l1: 0.0397 - val_loss: 0.0549 - val_pure_l1: 0.0411 -
learning_rate: 0.0010
Epoch 22/30
235/235 68s 289ms/step -
loss: 0.0737 - pure_l1: 0.0399 - val_loss: 0.1360 - val_pure_l1: 0.0496 -
learning_rate: 0.0010
Epoch 23/30
235/235 83s 353ms/step -
loss: 0.0731 - pure_l1: 0.0395 - val_loss: 0.0568 - val_pure_l1: 0.0400 -
learning_rate: 0.0010
Epoch 24/30
235/235 81s 346ms/step -
loss: 0.0770 - pure_l1: 0.0411 - val_loss: 0.0541 - val_pure_l1: 0.0369 -
learning_rate: 0.0010
Epoch 25/30
235/235 70s 301ms/step -
loss: 0.0761 - pure_l1: 0.0403 - val_loss: 0.0503 - val_pure_l1: 0.0335 -
learning_rate: 0.0010
Epoch 26/30
235/235 70s 299ms/step -
loss: 0.0736 - pure_l1: 0.0393 - val_loss: 0.3690 - val_pure_l1: 0.1528 -
learning_rate: 0.0010
Epoch 27/30
235/235 70s 299ms/step -
loss: 0.0728 - pure_l1: 0.0397 - val_loss: 0.0592 - val_pure_l1: 0.0413 -
learning_rate: 0.0010

Epoch 28/30

235/235 71s 301ms/step -

loss: 0.0718 - pure_l1: 0.0387 - val_loss: 0.0585 - val_pure_l1: 0.0405 -

learning_rate: 0.0010

Epoch 29/30

235/235 70s 297ms/step -

loss: 0.0757 - pure_l1: 0.0403 - val_loss: 0.0507 - val_pure_l1: 0.0345 -

learning_rate: 0.0010

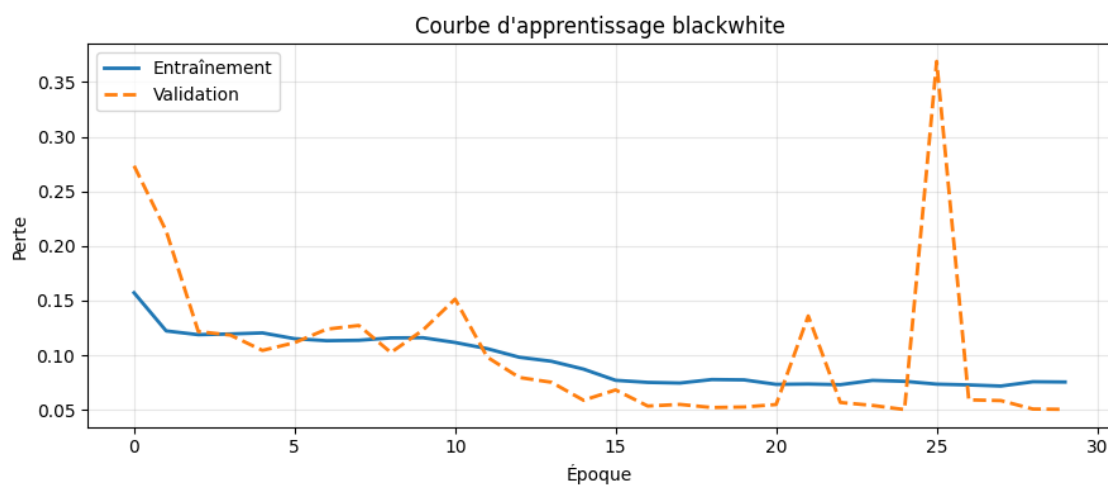
Epoch 30/30

235/235 69s 293ms/step -

loss: 0.0755 - pure_l1: 0.0402 - val_loss: 0.0504 - val_pure_l1: 0.0349 -

learning_rate: 0.0010

Restoring model weights from the end of the best epoch: 25.



```
[29]: print(os.listdir('./aeplus_models/'))
```

```
['unet_bw_combined.keras', 'unet_bw_combined_2.keras',  
'unet_bw_combined_3.keras', 'checkpoint_epoch01.keras',  
'checkpoint_epoch02.keras', 'checkpoint_epoch03.keras',  
'checkpoint_epoch04.keras', 'checkpoint_epoch05.keras',  
'checkpoint_epoch09.keras', 'checkpoint_epoch12.keras',  
'checkpoint_epoch13.keras', 'checkpoint_epoch15.keras',  
'checkpoint_epoch25.keras']
```

```
[30]: from skimage.metrics import peak_signal_noise_ratio as compute_psnr  
      from skimage.metrics import structural_similarity as compute_ssim  
  
      def show_reconstruction(model, typeimage, n=6, dataset_dirs=COMBINED_DIRS):  
          """  
          Affiche n triplets : (corrompue / reconstruite / originale)  
          avec scores PSNR et SSIM par image.
```

```

"""
ds = make_tf_dataset(dataset_dirs, TARGET_SIZE, batch_size=n,
                      typeimage=typeimage, use_augment=False)
corr_batch, orig_batch = next(iter(ds))
corr = corr_batch.numpy()
orig = orig_batch.numpy()
recon = model.predict(corr_batch, verbose=0)
fig, axes = plt.subplots(3, n, figsize=(3.2 * n, 10))

row_labels = {
    0: 'N&B (entrée)' if typeimage == 'blackwhite' else f'{typeimage}_
↳(entrée)',
    1: 'Reconstruction',
    2: 'Originale',
}
cmap_input = 'gray' if typeimage == 'blackwhite' else None

for i in range(n):
    orig_u8 = (np.clip(orig[i], 0, 1) * 255).astype(np.uint8)
    recon_u8 = (np.clip(recon[i], 0, 1) * 255).astype(np.uint8)

    psnr = compute_psnr(orig_u8, recon_u8, data_range=255)
    ssim = compute_ssim(orig_u8, recon_u8,
                        data_range=255, channel_axis=-1)

    psnr_color = 'green' if psnr > 25 else ('orange' if psnr > 20 else
↳'firebrick')

    # Ligne 0 : image corrompue
    axes[0, i].imshow(np.clip(corr[i], 0, 1), cmap=cmap_input)
    axes[0, i].set_title(f'Sample {i+1}', fontsize=9)
    axes[0, i].axis('off')

    # Ligne 1 : reconstruction
    axes[1, i].imshow(np.clip(recon[i], 0, 1))
    axes[1, i].set_xlabel(
        f'PSNR {psnr:.1f} dB | SSIM {ssim:.3f}',
        color=psnr_color, fontsize=8
    )
    axes[1, i].axis('off')

    # Ligne 2 : originale
    axes[2, i].imshow(np.clip(orig[i], 0, 1))
    axes[2, i].axis('off')

# Labels des lignes
for row, label in row_labels.items():

```

```

axes[row, 0].set_ylabel(label, fontsize=11, labelpad=8)

plt.suptitle(f'{typeimage} - U-Net (best epoch sauvegardé)', fontsize=12,
↳y=1.01)
plt.tight_layout()
plt.show()

psnr_scores, ssim_scores = [], []
for i in range(n):
    o = (np.clip(orig[i], 0, 1) * 255).astype(np.uint8)
    r = (np.clip(recon[i], 0, 1) * 255).astype(np.uint8)
    psnr_scores.append(compute_psnr(o, r, data_range=255))
    ssim_scores.append(compute_ssim(o, r, data_range=255, channel_axis=-1))

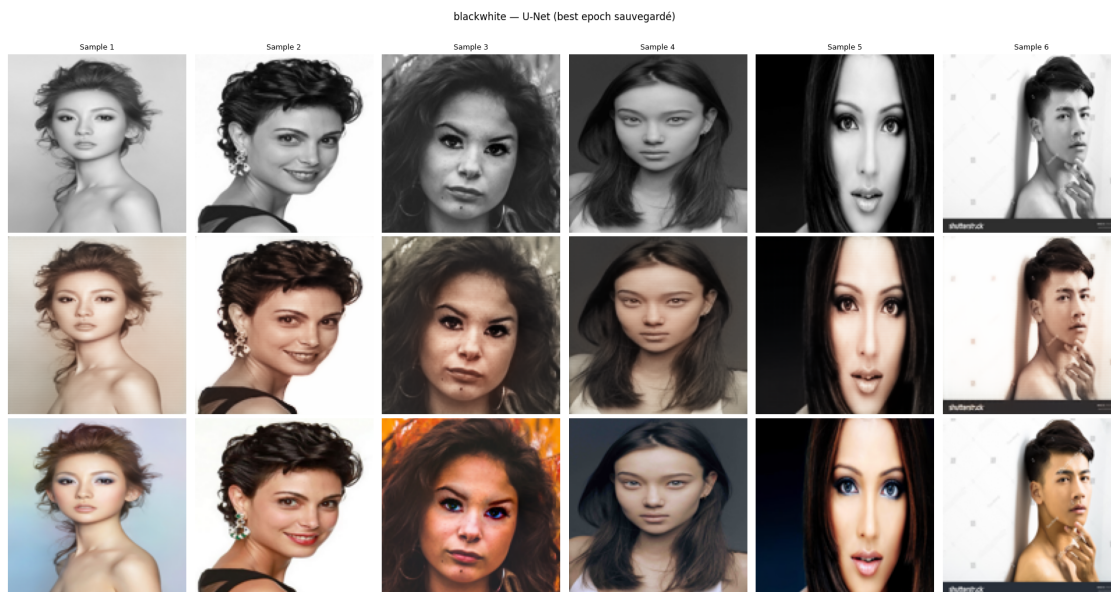
print(f"\n{' '*40}")
print(f"  PSNR moyen : {np.mean(psnr_scores):.2f} dB  (±{np.
↳std(psnr_scores):.2f})")
print(f"  SSIM moyen : {np.mean(ssim_scores):.4f}  (±{np.std(ssim_scores):.
↳4f})")
print(f"{' '*40}\n")

model_bw = tf.keras.models.load_model(
    './aeplus_models/checkpoint_epoch25.keras',
    compile=False
)

show_reconstruction(model_bw, 'blackwhite', n=6)

```

[make_tf_dataset] 18834 images (blackwhite, augment=False)



PSNR moyen : 27.12 dB (± 3.90)
SSIM moyen : 0.9497 (± 0.0333)

[]: